

What is optimal
on a martingale?

theory Prop. 1



excess risk = mean
squared departure

What do models
forecast on the null?

null ladder

N1 random walk

N2 GARCH

N3 heavy tails

N4 bid-ask bounce

optimum known on each

Where does the
direction come from?

controls

level, sign

frequency

model size

context

released
model

one input at a time

Is skill on prices
more than the bias?

instrument



randomise
return signs



split into even
and odd parts

Can the bias be
removed safely?

theory Prop. 2

gain guaranteed
under symmetry

otherwise an exact
split of the risk

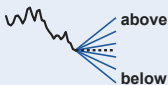
instrument: mirror



even $\frac{1}{2}(D(x) + D(-x))$
same for x and -x

odd $\frac{1}{2}(D(x) - D(-x))$
flips sign with x

experiment



11 frozen models,
horizons 1 to 128

experiment: corpus



growth

encoder
decoder



symmetric

encoder
decoder

only the corpus differs

experiment



falling markets,
predicted first



timing and
calendar shift

experiment

keep the odd part
at inference



fine-tune on
x and -x